

Equivalent Ratios

Lesson 3-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*3 red apples : 5 green apples***Ratio**

A way to compare two amounts.

2:3 and 4:6 both mean 2 out of every 3, so $\frac{2}{3} = \frac{4}{6}$ **Equivalent ratios**

Two ratios that mean the same thing.

*60 miles per 1 hour***Rate**

A ratio comparing two amounts with different units, like miles per hour.

 $\frac{2}{3} = \frac{4}{6}$ **Proportion**

A math sentence saying two ratios are equal.

 $12:18 \rightarrow$ divide both by 6 $\rightarrow 2:3$ **Simplify**

To make a ratio smaller while keeping the same comparison.

Key Ideas & Notes

- Chef Montoya is preparing her famous tomato soup for a school banquet.
- Her original recipe serves 8 people, but tonight she needs to serve 32 guests.
- She needs your help to scale the recipe without changing the flavor!
- Chef Montoya's recipe calls for 3 cups of tomatoes for every 2 cups of broth. Complete the ratio table to help her scale the recipe.

Think About It

- What quantities are being compared in the recipe?
- What stays the same when you multiply both ingredients by the same number?
- How many times bigger is 32 than 8?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Montoya's soup serves 8 people, but she needs to serve 32. How many times bigger is the recipe, and what must she do to BOTH ingredients to keep the flavor?

Level 1 support

Start here: 32 is 4 times 8, so she multiplies both ingredients by 4.

32 is ____ times bigger than 8, so I multiply both by ____.

32 es ____ veces más grande que 8, así que multiplico ambos por ____.

The flavor stays the same because ____.

El sabor se mantiene igual porque ____.

Word bank: **Equivalent ratios** **Proportion** **Rate**

Listen for: A strong answer finds the scale factor 4 ($32 \div 8$) and explains both ingredients must be multiplied by 4 to make equivalent ratios.

Level 2

Chef Montoya now needs to scale DOWN to serve only 4 people. Explain what scale factor she uses and why dividing keeps the ratios equivalent.

- To serve 4 people she divides both ingredients by ____ because ____.
- Dividing keeps equivalent ratios because ____.

EXPLORE

Looking at the tomatoes-to-broth table (3:2, 6:4, 9:6, 12:8), how can you PROVE these ratios are all equivalent?

Level 1 support

Start here: Each row is 3:2 multiplied by the same number, so they all simplify to 3:2.

These ratios are equivalent because each is ____ times ____.

Estas razones son equivalentes porque cada una es ____ veces ____.

I checked by ____ both numbers.

Lo comprobé al ____ ambos números.

Word bank: Equivalent ratios Simplify Proportion Rate

Listen for: Listen for proof by simplifying (all reduce to 3:2) or scaling (each multiplied by the same factor), not just 'they look similar.'

Level 2

A student says 6:4 and 9:6 are equivalent because 'you add 3 and 2 each time.' Critique this reasoning even though the rows happen to be correct.

- Adding 3 and 2 makes correct rows here, but it does NOT prove ____.
- The real reason they are equivalent is that they ____ to the same ratio.

CONNECT

The smoothie shop uses 2 cups of strawberries for every 3 cups of yogurt and needs 15 cups of yogurt. How is this the same as scaling Chef Montoya's soup?

Level 1 support

Start here: Both scale a 2-quantity recipe with equivalent ratios.

This is like our ratio work because ____ and ____ are related by ____.

Esto es como nuestro trabajo con razones porque ____ y ____ se relacionan por ____.

15 cups of yogurt is ____ times 3, so I need ____ cups of strawberries.

15 tazas de yogur son ____ veces 3, así que necesito ____ tazas de fresas.

Word bank: **Equivalent ratios** **Proportion** **Simplify** **Rate**

Listen for: A strong answer uses the scale factor 5 ($15 \div 3$) to find 10 cups of strawberries and names this as making an equivalent ratio / proportion.

Level 2

Set up this smoothie situation as a proportion ($2/3 = ?/15$) and explain how cross-multiplying confirms your strawberry amount.

- The proportion is $2/3 = ___ / 15$, and cross-multiplying gives ____.
- Cross-multiplying works because equivalent ratios always ____.

Guided Examples

Example 1

Which ratio is equivalent to 2:5?

Solution: Multiply both parts by 2: $2 \times 2 = 4$ and $5 \times 2 = 10$, so 4:10 is equivalent to 2:5.

Answer: A. 4:10

Example 2

A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

Solution: Divide both by 2: $4 \div 2 = 2$ and $6 \div 2 = 3$. The ratio 2:3 is equivalent to 4:6.

Answer: A. 2:3

Example 3

Which ratio is NOT equivalent to 6:9?

Solution: 6:9 simplifies to 2:3. Check each: 2:3 ✓, $12:18 = 2:3$ ✓, $18:27 = 2:3$ ✓. But 3:5 does not simplify to 2:3.

Answer: A. 3:5

Write About the Math

The Writing Revolution

I can explain my reasoning using the words equivalent ratios, rate, proportion, and simplify.

1. Kernel Sentence subject + verb

Model: Ratio is a way to compare two amounts.

Razón es una manera de comparar dos cantidades.

Write a kernel sentence about ratio. Use a subject and a verb.

Escribe una oración base sobre razón. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Ratio matters in math

Razón importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Ratio matters in math because ____.

Razón importa en matemáticas porque ____.

but
pero

Ratio matters in math, but ____.

Razón importa en matemáticas, pero ____.

so
entonces

Ratio matters in math, so ____.

Razón importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about ratio.
Di un hecho verdadero sobre ratio.

Ratio ____.

Question
Pregunta

Ask a question about ratio.
Haz una pregunta sobre ratio.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about ratio.
Muestra entusiasmo sobre ratio.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with ratio.
Dile a un compañero qué hacer con ratio.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

32 is ____ **times bigger than 8, so I multiply both by** ____.

32 es ____ *veces más grande que 8, así que multiplico ambos por* ____.

The flavor stays the same because ____.

El sabor se mantiene igual porque ____.

These ratios are equivalent because each is ____ **times** ____.

Estas razones son equivalentes porque cada una es ____ *veces* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Marcus earns \$45 for every 3 hours of work. How much does he earn in 8 hours?

- A. \$120
- B. \$90
- C. \$135
- D. \$105

Show your work:

2. A store sells 3 pens for \$2. Another store sells 9 pens for \$6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Marcus's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A store sells 5 notebooks for \$8. At this rate, how much would 15 notebooks cost?

- A. \$24
- B. \$18
- C. \$20
- D. \$40

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. \$120 — *Unit rate: $\$45 \div 3 = \$15/\text{hour}$. In 8 hours: $\$15 \times 8 = \120 .*
2. **Try It 2:** Yes, they are equivalent. Method 1 (simplify): $9:6 \div 3 = 3:2$, which matches 3:2. Method 2 (cross-multiply): $3 \times 6 = 18$ and $2 \times 9 = 18$. Since both cross products are equal, the ratios are equivalent.
3. **Exit Ticket:** A. \$24 — *15 is 3 times 5, so the cost is $3 \times \$8 = \24 . The ratio 5:8 scales to 15:24.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Ratio is a way to compare two amounts.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).